

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	SWUID2026217449
Project Name	Rising Waters - Flood Prediction ML
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Weather Data Input Form	Form to input rainfall, cloud visibility, temperature, humidity, seasonal rainfall	Frontend (UI)	Done	1
2	Data Preprocessing	Cleans and scales input data using StandardScaler	Backend / Logic	Done	1
3	Decision Tree Model	Baseline classification model	ML Model	Done	1
4	Random Forest Model	Best performing model (75.20% accuracy)	ML Model	Done	1
5	KNN Model	Comparison classification model	ML Model	Done	1
6	XGBoost Model	Comparison classification model	ML Model	Done	1
7	Prediction Result Display	Displays High/Low flood risk result on web page	Frontend (UI)	Done	1
8	Model Persistence (.pkl)	Saves trained model and scaler for reuse	Backend / Logic	Done	1

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2 (Multi-model comparison, Model persistence)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input form, Result display
2	Backend / Logic	2	Data preprocessing, Model persistence
3	Database / Storage	0	N/A - uses flat file dataset
4	API / Integration	1	Flask REST-style route for prediction
5	Security / Authentication	0	N/A - single user local app